



0.65 MILES THE LENGTH OF THE TUNNEL OF EUPALINOS AT SAMOS

The Tunnel of Eupalinos, built in the 6th century BCE, may have been excavated accurately by surveying a series of right-angled triangles above ground.

Abdera (460–370 BCE), who postulated that matter was made up of an infinite number of minute, indivisible particles.

The most famous mathematician of the ancient world was the Greek, **Pythagoras of Samos** (c.580–500 BCE). He established a school that promoted the **mystical powers of numbers** and particularly of the *tetraktys*, the perfect arrangement of 10 as a triangle of four rows. He is best known for the theorem bearing his name (see panel, below), but he also firmly believed in the **transmigration of souls** and his followers lived by a strict set of rules, including a prohibition on eating beans.

The oldest major Chinese mathematical treatise, the

PLATO (424–348 BCE)

One of the most influential of the ancient Greek philosophers, Plato proposed a type of ideal society ruled by philosopher-kings, and espoused the importance of ethics as a guide to a just life. In his many works, he set out a theory of ideal “forms,” of which the material world is only a reflection. Most of his books are cast in the form of dialogues by his teacher Socrates.



Zhou Bi Suan Jing (some parts of which date to as early as 500 BCE), contain proof of Pythagoras’s theorem. At about the same time, or possibly

earlier, **Chinese mathematicians also invented magic squares**—square grids of numbers in which the numbers in all rows, all columns, and both diagonals add up to the same total.

By around 530 BCE, Greek surveying expertise had advanced sufficiently to allow the engineer **Eupalinos of Samos** to excavate a **water channel** 0.65 miles (1.04 km) through a hillside by digging tunnels from each end. The two tunnels met almost perfectly in the middle. Eupalinos may have used Pythagoras’s theorem to survey right-angled triangles above ground to determine the path of the channel.

Indian astronomy is thought to have its roots in the Indus Civilization. Ancient Hindu sacred

scriptures called the *Vedas*—completed c.500 BCE—contain references to using astronomical observations for calculating the dates of religious ceremonies and identifies **28 star patterns in the night sky** to help track the movements of the Moon.

In the 5th century BCE, Greek thinkers moved away from simple cosmological theories toward more sophisticated ideas about **the nature of the universe**. Heraclitus (c.535–475 BCE) sought to explain phenomena in terms of flux and change. He also believed in the unity of opposites, saying “the road is the same both up and down.” **Empedocles of Acragas** (494–434 BCE) believed that all matter consisted of varying proportions of earth, air, fire, and water. This **theory of four elements** remained influential for many centuries.

Mesoamerican calendrics

This Zapotec stele from Monte Albán in Mexico dates from 500–400 BCE and contains some of the earliest calendary glyphs from Mesoamerica.

The Maya of Mesoamerica developed a complex **calendrical system** based on a series of cycles based on the number 20, which may initially have been developed by the Olmecs (the first major civilization in Mexico) before the 5th century BCE. The Mayan Haab (year) had 18 months of 20 days plus one of five days—**one of the two elements of the Calendar Round cycle**. Mayan astronomers also oriented monuments to sunset positions at the equinoxes and solstices, and were able to predict eclipses.

The cities of the Indus Valley were laid out in a grid pattern around 2600 BCE, but the first person to theorize **urban planning** was Hippodamus of Miletus (493–408 BCE). He is said to have devised an ideal city for 10,000 citizens, laid out on a grid. Using his “Hippodamian grid,” he also laid out Piraeus, the harbour town of Athens, and Thurii in Italy.



glyph for Zapotec year “Four Serpent”

glyph for Zapotec day “Eight Water”

PYTHAGORAS’S THEOREM

The theorem of Pythagoras states that the sum of the squares of the two short sides of a right-angled triangle are equal to the square of the hypotenuse (the long side). Although associated with the Greek mathematician Pythagoras, the theorem was known to the Babylonians around 1800 BCE and possibly also to the Egyptians as early as 1900 BCE.

